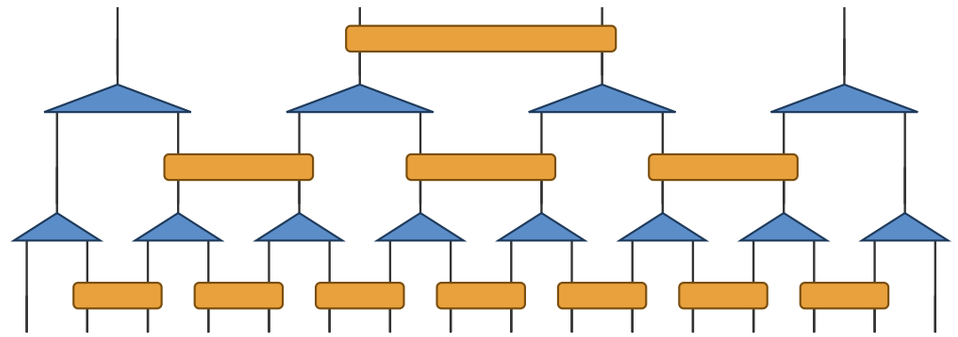
Tensor networks methods for low-dimensional quantum systems



Lecture 0: info on the course

UNIVERSITY OF WARSAW

WYDZIAŁ FIZYKI

Schedule

Lectures:

1) **quantum many-body physics**
2) **entanglement entropy**
3) **tensor networks - Matrix product states**
4) **density matrix renormalization group (DMRG)**
5) **non-equilibrium dynamics and quantum circuits**

Workshops:

1) **exact diagonalization**
2) **phase diagrams from tensor networks**
3) **dynamics with tensor networks**

Students' projects:

1) **small research projects**
2) **from the beginning of November**
3) **talks in January**

Format and grades

Wednesday 09:15 - 13:00
room 1.30

Format varies from week to week:
1) lectures + tutorials
2) lectures + workshops
3) workshops

Lectures + tutorials
Miłosz Panfil
office 4.12
milosz.panfil@fuw.edu.pl

Workshops
Jędrzej Wardyn
office 4.29
jedrzej.wardyn@fuw.edu.pl

Grades:
- **Midterm exam 18.11 (25 %)**
- **Presentations 13.01 or 20.01 (50 %)**
- **Final exam 27.01 (25 %)**

Numerics







We will use Julia + XDiag + iTensors

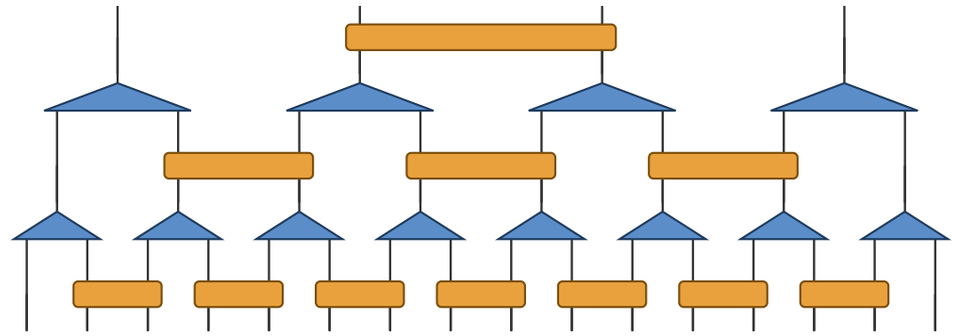
By 21.10 you need to have Julia + XDiag running

GitHub repository: https://github.com/mkpanfil/TN_UW

- scripts

- extra materials

Tensor networks methods for low-dimensional quantum systems



Lecture 1:
Quantum many-body physics (in 1d)

UNIVERSITY OF WARSAW

WYDZIAŁ FIZYKI

Quantum mechanics

states: elements of a Hilbert space

$$|a\rangle \in \mathscr{H}$$

wavefunction -
components of the vector in some basis

$$\psi(x) = \langle x|a\rangle$$

product structure - local Hilbert spaces $\mathscr{H} = \otimes_{j=1}^{N} \mathfrak{h}_j$ $\mathfrak{h}_j = \mathfrak{h}_k$

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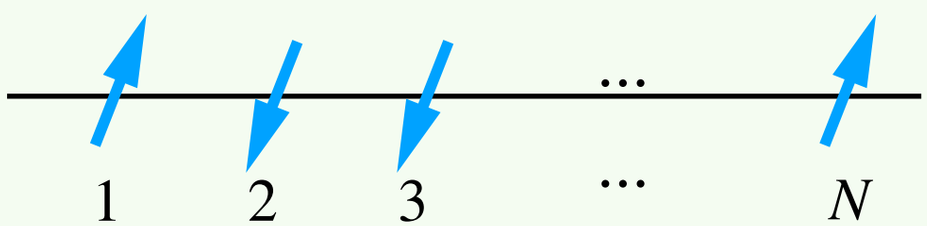
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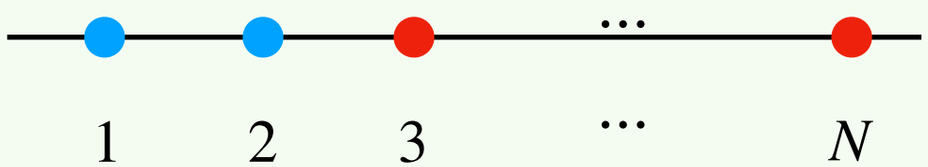
Examples

spin chains



$\mathfrak{h}_j = \mathbf{C}^2$ $|a\rangle = \alpha|\uparrow\rangle + \beta|\downarrow\rangle$

lattice gas of spinless fermions



$\mathfrak{h}_j = \mathbf{C}^2$ $|a\rangle = \alpha|1\rangle + \beta|0\rangle$

Hamiltonians

Operators $\hat{\mathcal{O}} : \mathscr{H} \to \mathscr{H}$

Physics -> Hamiltonian $\hat{H}$

Local operators $\hat{o}_j : \mathfrak{h} \to \mathfrak{h}$

$$\hat{o}_j = \mathbf{1} \otimes \ldots \otimes \mathbf{1} \otimes \hat{o} \otimes \mathbf{1} \otimes \ldots \otimes \mathbf{1}$$

Hamiltonians

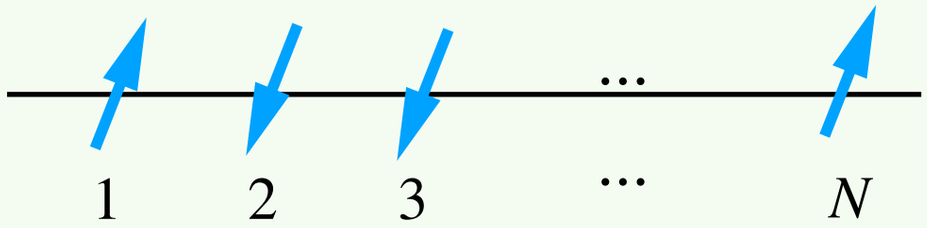
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spin chains



Ising model

$$\hat{H} = \sum_{j=1}^{N} \left(s_j^x s_{j+1}^x + h s_j^z \right)$$

Heisenberg model

$$\hat{H} = \sum_{j=1}^{N} \left(s_j^x s_{j+1}^x + s_j^y s_{j+1}^y + s_j^z s_{j+1}^z \right)$$

theory of magnetism
quantum phase transitions

Hamiltonians

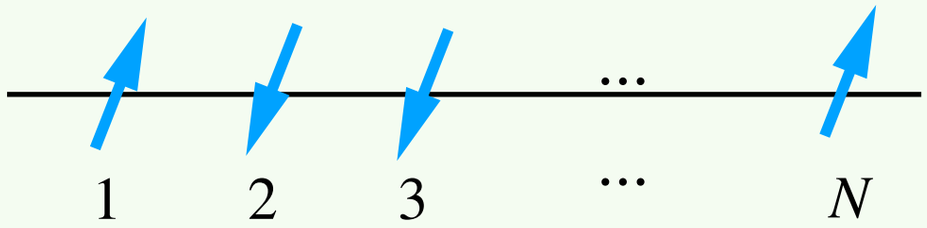
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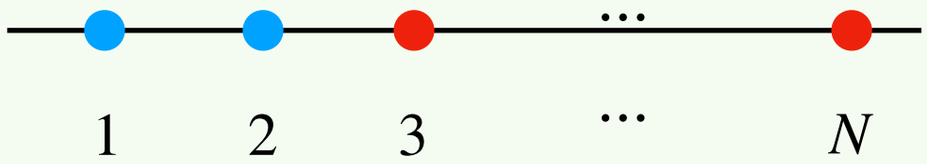
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theory of magnetism
quantum phase transitions

lattice gas of spinless fermions



lattice gas

$$\hat{H} = -t \sum_{j=1}^{N} \left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j \right) + U \sum_{j=1}^{N} n_j n_{j+1}$$

hopping (kinetic) **interaction**

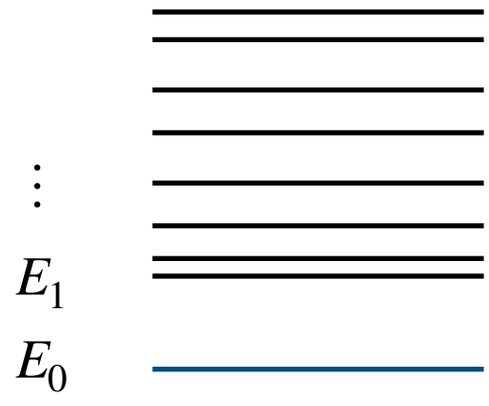
$$n_j = c_j^\dagger c_j$$

electrons in crystals
theory of (super)conductivity

Eigenstates and the ground states

Solve the eigenproblem = diagonalize the Hamiltonian

$$\hat{H}\,|\psi\rangle = E_\psi\,|\psi\rangle$$



At low temperatures physics dominated by the ground state and low-lying excited states

$$e^{-\beta\hat{H}} = e^{-\beta E_0} \times e^{-\beta(\hat{H}-E_0)}$$

gives zero for $E_1 - E_0 > k_B T$

Task 1: Find the ground state of a given Hamiltonian

Possible solution

1. **Diagonalise the Hamiltonian**
2. **Identify the lowest lying state(s)**

Spin algebra

Spin algebra on a single site

$$[\hat{s}^{\alpha}, \hat{s}^{\beta}] = i\epsilon_{\alpha,\beta,\gamma}\hat{s}^{\gamma}$$

Spin algebra on different sites

$$[\hat{s}_j^{\alpha}, \hat{s}_k^{\beta}] = 0, \quad \text{for } j \neq k$$

Casimir operator

$$\hat{C} = \left(\hat{s}^x\right)^2 + \left(\hat{s}^y\right)^2 + \left(\hat{s}^z\right)^2$$

$$[\hat{C}, \hat{s}^{\alpha}] = 0$$

spin raising and lowering operator

$$\hat{s}^{\pm} = \hat{s}^x \pm i\hat{s}^y$$

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spin raising and lowering operator

$$\hat{s}^{\pm} = \hat{s}^{x} \pm i\hat{s}^{y}$$

two dimensional representation - spin 1/2

$$|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\hat{s}^{x} = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \hat{s}^{y} = \frac{1}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \hat{s}^{z} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$s^{z}|\uparrow\rangle = \frac{1}{2}|\uparrow\rangle \qquad s^{z}|\downarrow\rangle = -\frac{1}{2}|\downarrow\rangle$$

$$s^{-}|\uparrow\rangle = |\downarrow\rangle \qquad s^{+}|\downarrow\rangle = |\uparrow\rangle$$

$$\hat{C} = \frac{3}{2}\hat{1} \qquad (s^{-})^2 = 0 \qquad (s^{+})^2 = 0$$

Heisenberg chain with 2 sites

$$\hat{H} = J(\hat{s}_1^x \hat{s}_2^x + \hat{s}_1^y \hat{s}_2^y + \Delta \hat{s}_1^z \hat{s}_2^z)$$

use spin raising and lowering operators

$$\hat{s}^{\pm} = \hat{s}^x \pm i\hat{s}^y$$

the Hamiltonian becomes

$$\hat{H} = \frac{1}{2} J \left(\hat{s}_1^+ \hat{s}_2^- + \hat{s}_1^- \hat{s}_2^+ \right) + J\Delta \hat{s}_1^z \hat{s}_2^z$$

choose a basis

$$|0\rangle = |\downarrow\downarrow\rangle \qquad |1\rangle = |\downarrow\uparrow\rangle$$

$$|2\rangle = |\uparrow\downarrow\rangle \qquad |3\rangle = |\uparrow\uparrow\rangle$$

action of the Hamiltonian

$$\hat{H}|0\rangle = \frac{J\Delta}{4}|0\rangle \qquad \hat{H}|1\rangle = -\frac{J\Delta}{4}|1\rangle + \frac{J}{2}|2\rangle$$

$$\hat{H}|2\rangle = \frac{J}{2}|1\rangle - \frac{J\Delta}{4}|2\rangle \qquad \hat{H}|3\rangle = \frac{J\Delta}{4}|3\rangle$$

Heisenberg chain with 2 sites

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Hamiltonian matrix

$$\langle a|\hat{H}|b\rangle = \frac{J}{4}\begin{pmatrix} \Delta & 0 & 0 & 0 \\ 0 & -\Delta & 2 & 0 \\ 0 & 2 & -\Delta & 0 \\ 0 & 0 & 0 & \Delta \end{pmatrix}$$

Eigenstates

$$|0\rangle = |\downarrow\downarrow\rangle \qquad |3\rangle = |\uparrow\uparrow\rangle \qquad E = \frac{J\Delta}{4}$$

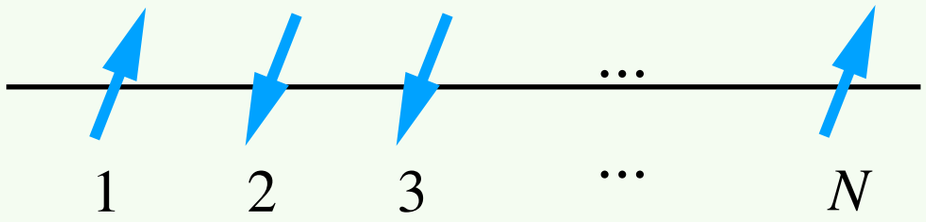
$$|s\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle\right) \qquad E = -\frac{J}{4}(\Delta + 2)$$

$$|t\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle\right) \qquad E = -\frac{J}{4}(\Delta - 2)$$

Quantum phase transitions

Ground state dependence of the Hamiltonian parameters

spin chains



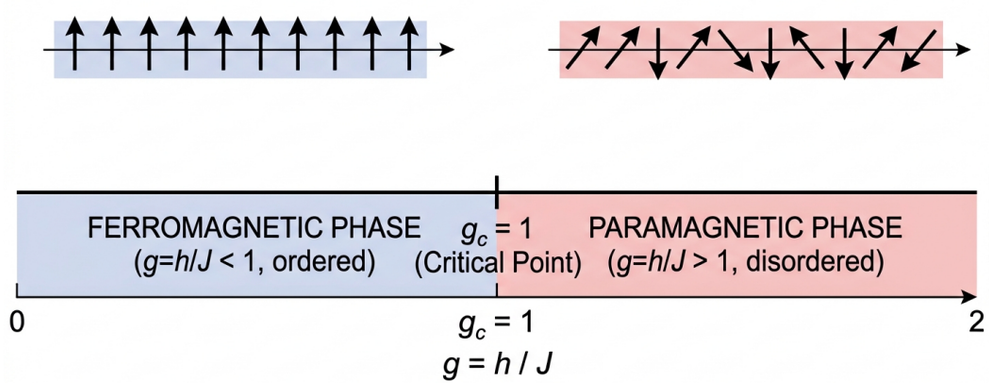
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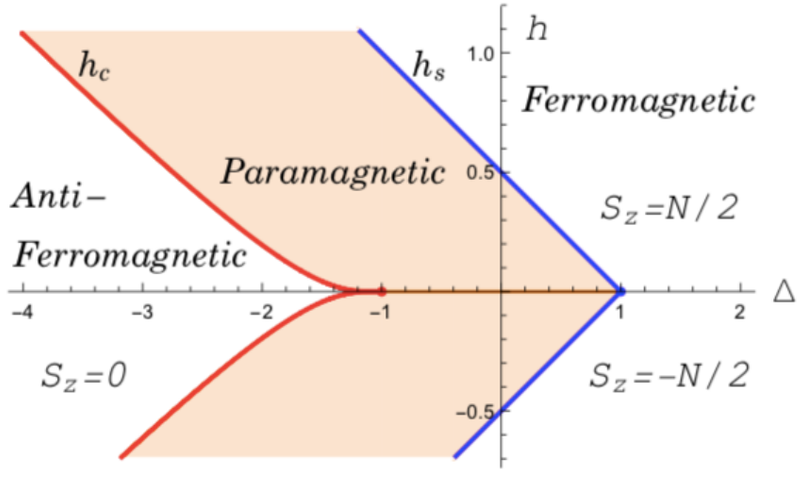
XXZ model [a generalization of Heisenberg (XXX)]

$$\hat{H} = \sum_{j=1}^{N} \left(s_j^x s_{j+1}^x + s_j^y s_{j+1}^y + \Delta s_j^z s_{j+1}^z \right)$$

Phase diagram Ising



Phase diagram XXZ



Symmetries

Symmetries represented by operators U such that

- $[H, U] = 0$ **or** $U^\dagger H U = H$
- U **is unitary (or antiunitary)**

Noether theorem:
continuous symmetry $U(\alpha) = e^{-i\alpha Q}$ **generated by** Q **then** Q **is conserved:** $[H, Q]$

Degeneracies in the spectrum:
eigenstates $|\psi\rangle$ **and** $U|\psi\rangle$ **have the same energies**

Let $H|\psi\rangle = E|\psi\rangle$ **then** $H(U|\psi\rangle) = UH|\psi\rangle = E(U|\psi\rangle)$

Because $[H, U] = 0$**, the Hilbert space decomposes into subspaces** $\mathscr{H} = \bigoplus_r \mathscr{H}_r$

Each subspace can be diagonalised independently